Breathing in the Classroom

Pupil sheet

Name: _____ **Date:** _____

Safety: Only the teacher opens the window and the door; nobody breathes over the box or sprays anything near it.

1. What I see on the screen

On the start page, our box sees: ☐ CO_2 (SCD41) ☐ VOC ☐ temperature and humidity (BME280)

Three rules. I stay **at least a metre** from the box. The recorder presses **Window open**, **Door open**, **Window closed** or **Door closed at the moment** it happens. The box writes one row every **20 seconds**; I do not stop the recording before the bell.

2. What I do on my own

First I predict. I do not change my prediction after I measure.
I sketch the CO_2 during the lesson, with the moment the window opens:

CO_2 (ppm) ↑

time →

At the start I expect about _____ ppm, at most _____ ppm.

The event log (kept by the recorder).

Time	What happened	CO_2 on the card (ppm)
	Start recording, door and window closed	
	Stop recording	

How many of us were in the room: _____ We aired with: ☐ the window ☐ the door ☐ both

In Excel (lesson 2), from the columns `Elapsed (s)` and `CO2 (ppm)`:

With the room closed the CO_2 rose by _____ ppm in _____ min, that is _____ ppm per minute.

When the window opened, the CO_2 was _____ ppm; the excess above 420 ppm was _____ ppm. It fell to half after _____ min.

3. The conclusion I write

I predicted that, and the data show that _____

_____

The fall after opening was fast at first and slower later because _____

_____

The humidity and the temperature changed like this: _____

_____

The limit of our measurement: why can the CO_2 in the room never fall below about 420 ppm?

_____

Exit ticket

A five-minute break: do you open the door or the window? Why?

_____